Lecture 4:
Principles of ERP systems

Business Applications (Betriebswirtschaftliche Anwendungen)

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Hochschule für Technik
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Recap last week

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Course content

1	Course Kickoff
2	Introduction to ERP systems
3	SAP ERP
4	Principles of ERP systems
5	Modern ERP concepts
6	Fundamentals of AI agents
7	AI agent ecosystem & Retro
8	AI agents in business workflows
9	Agentic ERP in S/4 HANA
10	OpenClaw & Test preparation
11	E-Test

Learning objectives

At the end of the lecture, the students are able to …

- explain the **value chain** concept and how it maps to **enterprise domains**
- compare **functional**, **divisional** and **matrix organization** forms
- describe how **SAP S/4HANA** maps **organizational structures** to system data
- understand key **ERP technical features** such as **in-memory persistency** and **analytics**

Agenda

1 **Value Chain & Enterprise Domains**

2 Organizational Forms

3 ERP Tech Features

Value chain maps enterprise processes to activities

Primary activities

- Inbound Logistics
- Manu-facturing
- Outbound Logistics
- Marketing
- Sales and Service

Support activities

- Enterprise Infrastructure
- Human Resource Management
- Technology Development
- Procurement

Key Insight: The value chain (Porter, 1985) identifies and creates strategic competitive advantages by distinguishing direct value-adding activities from enabling support functions

Enterprise activities structured in four domains

1

Products and Services

Develop

R&D/Engineering, Product Management

2

Customer

Generate demand

Marketing, Sales, Customer Service, Commerce

3

Supply

Fulfill demand

Supply Chain, Manufacturing, Sourcing

4

Corporate

Plan and manage the enterprise

Finance, HR, Indirect Procurement

Each domain with specific functional departments

1 Products and Services	**Product Management** (e.g., portfolio optimization, product lifecycle, pricing strategy) **R&D / Engineering** (e.g., product testing, prototyping, design changes)
2 Customer	**Marketing** (e.g., market research, advertising campaigns, brand management) **Sales** (e.g., customer acquisition, sales strategy, CRM) **Customer Service** (e.g., helpdesk, complaint handling, service-level agreements) **Omnichannel Commerce** (e.g., online store, retail POS, marketplace integration)
3 Supply	**Supply Chain Planning** (e.g., demand forecasting, procurement planning) **Sourcing and Procurement** (e.g., supplier evaluation, contract negotiation, purchasing) **Manufacturing** (e.g., production execution, quality control, shop-floor management) **Service Delivery** (e.g., field service, maintenance scheduling, SLA tracking) **Supply Chain Execution** (e.g., transportation management, warehouse operations, logistics) **Supply Chain Enablement** (e.g., network design, analytics, sustainability tracking)
4 Corporate	**Finance** (e.g., accounting, treasury, financial risk management) **Indirect Procurement** (e.g., cost reduction initiatives, vendor management) **Asset Management** (e.g., plant maintenance, depreciation) **Human Resources** (e.g., recruiting, training, payroll, compensation) **Corporate Strategy** (e.g., M&A, long-term planning, competitive analysis) **Corporate Operations** (e.g., IT management, legal, compliance, facilities)

Enterprises require both a structural and a process view

Organizational Structure

Static View — "Who does what?"

- Depicts the hierarchical structure of an enterprise
- Resolves total output into subtasks and positions
- Classified into:

1D Functional, Divisional

2D Matrix

vs.

Process Organization

Dynamic View — "How does work flow?"

- Focuses on workflows and business processes
- Cross-functional perspective across departments
- Modeled using:

EPC Event-driven Process Chains

BPMN Business Process Model

Agenda

1 Value Chain & Enterprise Domains

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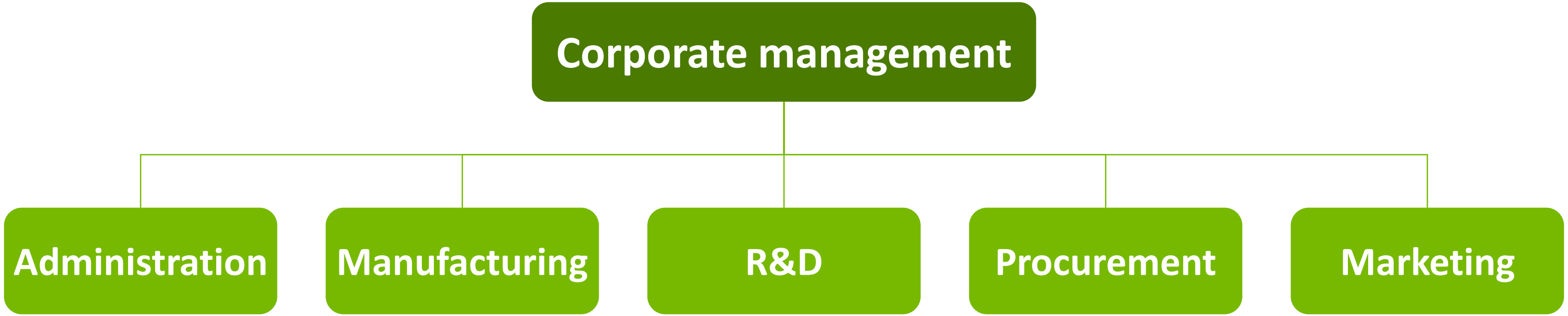
Functional organization groups by business function



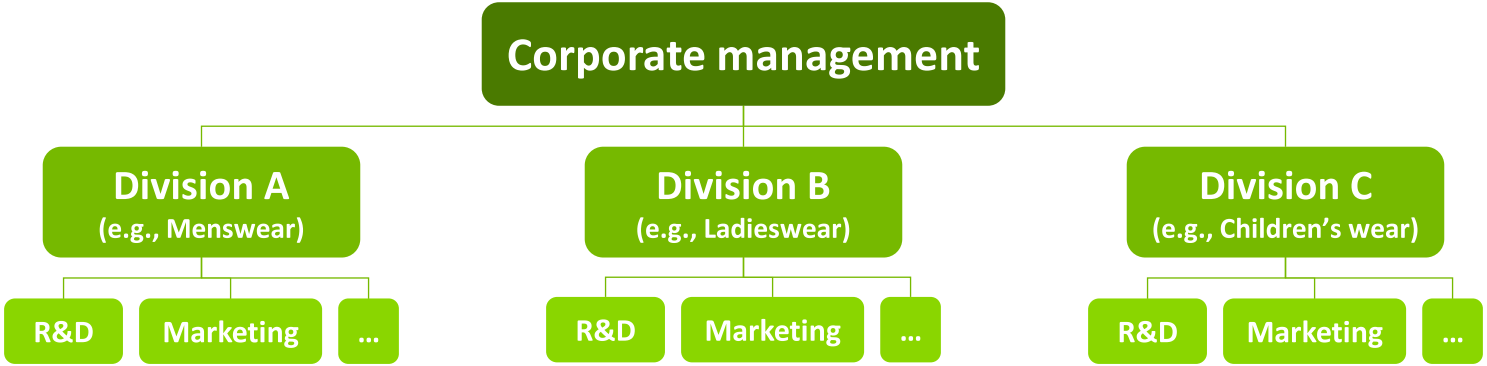
Characteristics

- Initial form of industrial enterprise
- Ideal for low product diversity
- Single design dimension

Structure

- Management board at top
- Operational units at lowest level
- Growth drives multi-level hierarchy

Divisional organization creates profit centers by object



Key Concept

- Autonomous business units
- By products, markets or geography
- Requires sufficient heterogeneity

Responsibility and Accountability

- Measured as profit centers
- Transfer prices between divisions
- Also configurable as cost centers

Matrix organization combines functional and object dimensions

Functional dimension →

Object dimension ↓

Corporate mgmt.	R&D	Marketing	…
Menswear	•	•	•
Womenswear	•	•	•
Children's wear	•	•	•

✓ Benefits

- Short communication paths
- Flexible market adjustment
- Diverse HR development chances

✗ Drawbacks

- Risk of competence conflicts
- Many trade-offs
- Requires highly qualified executives

Organizational data

Recap last week

Client *(Mandant)*
Highest level – represents entire SAP system

Company Code *(Buchungskreis)*
Legal entity for financial reporting

Plant *(Werk)*
Production or logistics site

Storage Location *(Lagerort)*
Physical warehouse location

Cross-Cutting Units

Sales Organization *(Vertriebsorg.)*
Sales structure for distribution

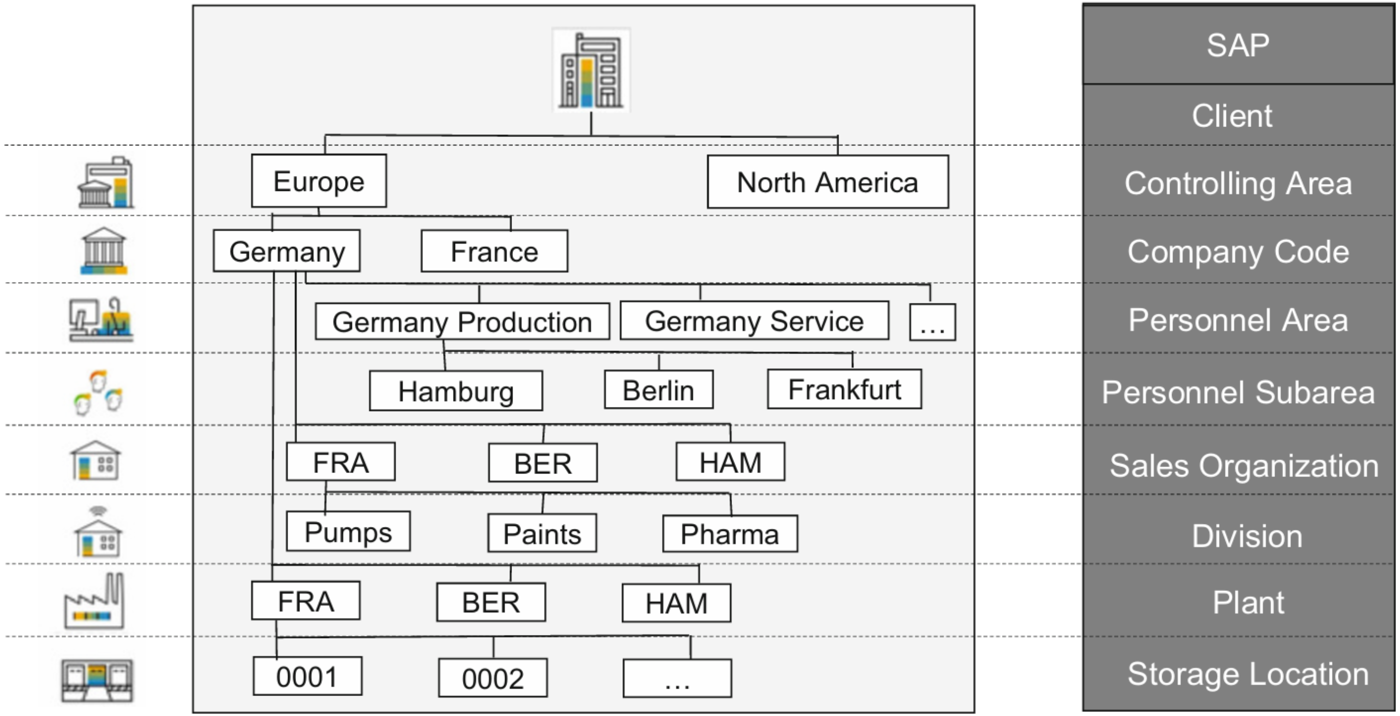
Division *(Sparte)*
Product line grouping

Business Area *(Geschäftsbereich)*
Internal reporting segment

SAP organizes enterprises into a hierarchy of organizational units. The client is the top-level container. Cross-cutting units like sales organization span multiple levels of the hierarchy.

Sources: SAP UCC Magdeburg (2025)

Exemplary enterprise structure



Objects in SAP S/4HANA Organizational Management

Purpose

- Component inside SAP S/4HANA HCM module
- Maintains master data for org structures using object-oriented data model

Key object types

Org Unit (key "O")
Functional unit of enterprise
(Dept, Group, Project Team)

Position (key "S")
Concrete position in the org
(e.g., Mechanic, Manager)

Person (key "P")
Employee holding a position
(e.g., Person Johnson)

Org plan shows relationships between objects

Example: SAP Org Plan

- PCAS AG (Root Org Unit)
  - Sales Dept (Org Unit)
    - Head of Sales (Position)
      - John Smith (Person)

Key Concepts

- **Org Plan:** Complete navigable hierarchy
- **Org Unit:** Departments, groups, project teams
- **Position:** Inherited from a job description
- **Person:** From HR master data

Leisure time

What content from this lecture can you recall so far?

Reflect together with the person sitting next to you

5 minutes

Agenda

1. Value Chain & Enterprise Domains
2. Organizational Forms
3. **ERP Tech Features**

Activity: Pick one topic from the following list, read the respective section in the Sarferaz book and present a short summary (3min) of the topic in front of class. Partner up in a group of 3-4 students.

Topics

- In-memory persistency (section 19)
- Virtual data model (section 20)
- Analytics (section 22)
- Search (section 23)
- In-app extensibility (section 28)
- Side-by-side extensibility (section 29)

Thank you for your attention

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